

CONFIDENTIAL DRAFT - STRICTLY FOR FACULTY REVIEW ONLY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

END-TERM THEORY EXAMINATION (DECEMBER 2026)

COURSE: CS302 - DESIGN AND ANALYSIS OF DEEP NEURAL NETWORKS

DURATION: 3 HOURS

MAXIMUM MARKS: 100

QUESTION 1: CONVOLUTIONAL NEURAL NETWORK DERIVATIONS (20 MARKS)

Problem Statement:

Consider an input image tensor X of size $64 \times 64 \times 3$ (Width $\times$ Height $\times$ Channels).

The input is passed through a Convolutional Layer (L1) with the following hyperparameters:

- Number of Filters (K): 32
- Filter / Kernel Size (F): 5×5
- Stride (S): 2
- Padding (P): 1
- Activation Function: ReLU

Questions:

- Calculate the exact output spatial dimensions (Width, Height, Depth) after passing through Layer L1. (5 Marks)
- Calculate the total number of trainable parameters (weights + biases) in Layer L1. (5 Marks)
- Derive the computational complexity in terms of total Floating Point Operations (FLOPs) required for the forward pass of Layer L1. (10 Marks)

Solution Key and Marking Scheme

a) Spatial Dimension Formula

$$W_{\text{out}} = \text{floor}((W_{\text{in}} - F + 2 \times P) / S) + 1$$

Substituting the given values:

$$W_{\text{out}} = \text{floor}((64 - 5 + 2 \times 1) / 2) + 1$$

$$= \text{floor}(61 / 2) + 1$$

$$= 30 + 1$$

$$= 31$$

Output Dimensions:

$$31 \times 31 \times 32$$

Marking:

- 3 Marks for correct formula application
- 2 Marks for exact tensor dimensions

b) Parameter Calculation

Each filter contains:

$$5 \times 5 \times 3 = 75 \text{ weights}$$

Each filter also has:

1 bias

Therefore,

$$\text{Parameters per filter} = 75 + 1 = 76$$

Total parameters for 32 filters:

$$32 \times 76 = 2,432 \text{ parameters}$$

Marking:

- 3 Marks for weight calculation
- 2 Marks for bias term addition

c) FLOPs Calculation

For each output unit:

Multiplications:

$$5 \times 5 \times 3 = 75$$

Additions:

75

Therefore,

Total FLOPs per output unit = 150

Total output units:

$$31 \times 31 \times 32 = 30,752$$

Total FLOPs:

$$30,752 \times 150 = 4,612,800 \text{ FLOPs}$$

$$\approx 4.61 \text{ MFLOPs}$$

Marking:

- 5 Marks for single-unit FLOP calculation
- 5 Marks for total FLOP aggregation

QUESTION 2: TRANSFORMER SELF-ATTENTION MECHANISM (20 MARKS)

Problem Statement:

Given an input sequence matrix X of dimension $(T \times d_{\text{model}})$, where:

- Sequence Length (T) = 4
- Embedding Dimension (d_{model}) = 8

Let W_Q , W_K , and W_V be projection weight matrices of dimension:

$$(d_{\text{model}} \times d_k)$$

where:

$$d_k = 4$$

Questions:

- Explain why the dot product ($Q \times K^T$) is scaled by $(1 / \sqrt{d_k})$ before applying the Softmax function. (10 Marks)
- Show mathematically how scaling prevents small gradients during backpropagation when d_k is large. (10 Marks)

Solution Key and Marking Scheme

a)

Assume that the query and key components are independent random variables with:

- Mean = 0
- Variance = 1

Their dot product has:

Mean = 0

Variance = d_k

As d_k increases, the dot-product values become larger in magnitude. Large values cause the Softmax function to saturate, producing probabilities close to 0 or 1. In these saturated regions, gradients become extremely small, leading to the vanishing gradient problem.

Dividing the dot product by $\sqrt{d_k}$ normalizes the variance back to approximately 1, keeping the Softmax outputs within a region where gradients remain useful for learning.

Marking:

- 5 Marks for variance derivation
- 5 Marks for explaining the relationship with Softmax saturation and vanishing gradients